



A comprehensive analysis of molecular characteristics of hot and cold tumor of gastric cancer

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Abstract

Background The advent of immunotherapy has revolutionized the treatment paradigm for gastric cancer (GC), offering unprecedented clinical benefits. However, a detailed molecular characterization of the tumor immune microenvironment in GC is essential to further optimize these therapies and enhance their efficacy.

Methods Consensus clustering was utilized to classify GC patients into distinct immune states, followed by an in-depth analysis of differences in mutation profiles, copy number variations, and DNA methylation patterns. Weighted gene co-expression network analysis (WGCNA) and correlation analysis were applied to identify gene modules underlying the classification of immune “hot” and “cold” tumors. Subsequently, 101 machine learning algorithm combinations were employed to construct a prognostic model based on the identified gene modules. Single-cell analysis was conducted to investigate cellular interactions associated with the immune-determinant gene module. Finally, immunofluorescence staining for CD8, CD45, and CXCR4 was performed on human GC tissue samples.

Results A total of 1,298 GC patients were included in this comprehensive analysis. For the first time, we identified and characterized immune “hot” and “cold” tumors in GC patients, revealing distinct molecular features associated with these tumor types. Immune “hot” tumor-related genes were identified, and their functional roles were validated through biological behavior analysis. A prognostic signature, termed the hot tumor top regulators (HTTR), was developed using 101 machine learning algorithm combinations. The HTTR signature emerged as an independent prognostic factor, effectively stratifying patients into low- and high-risk groups with significant differences in overall survival. High-risk groups demonstrated strong associations with immune checkpoint regulation, antigen presentation, and inhibitory pathways. Notably, single-cell analysis revealed that HTTR genes were highly active in CD8 + T cells, with the CXCL12-CXCR4 axis playing a critical role in mediating interactions between CD8 + T cells and endothelial cells.

Conclusion In conclusion, the HTTR signature served as a robust prognostic biomarker for GC patients and effectively identified those with immune “hot” tumors. This finding provided valuable insights into the molecular mechanisms of tumor immunity in GC, offering potential avenues for targeted therapeutic interventions.

Keywords Gastric cancer · Tumor immune microenvironment · Hot and cold tumors · Prognostic biomarker · CXCL12-CXCR4 axis

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Abbreviations

CAFs	Cancer-associated fibroblasts
CNV	Copy number variation
DMPs	Differential methylation positions
DMRs	Differential methylation regions
Enet	Elastic network
GBM	Gradient boosting machine
GC	Gastric cancer
GS	Gene significance
HTTR	Hot tumor top regulators
ICB	Immune checkpoint blockade
KEGG	Kyoto Encyclopedia of Genes and Genomes
MM	Module membership
OS	Overall survival
PAC	Proportion of ambiguous clustering
PCA	Principal component analysis
pIsRcox	Partial least squares regression for cox models and related techniques
RSF	Random survival forest
SSGSEA	Single-sample gene set enrichment analysis
STAD	Stomach adenocarcinoma
Super PC	Supervised principal component
Survivor-SVM	Survival support vector machine
TCGA	The cancer genome atlas
TEM	Effector memory T cells
TEX	Exhausted T cells
TME	Tumor microenvironment
TOM	Topological overlap matrix
TRM	Tissue-resident memory T cells
T-SNE	T-distributed stochastic neighbor embedding
WGCNA	Weighted correlation network analysis

Introduction

Gastric cancer (GC) remains one of the most prevalent malignancies globally, with over one million new cases and 769,000 deaths reported in 2020 [1]. The primary etiological factors include infections by *Helicobacter pylori* or Epstein–Barr virus, as well as lifestyle-related risk factors such as excessive alcohol consumption and high intake of salty foods [2, 3]. While surgical resection continues to be the cornerstone of curative treatment, chemotherapy and targeted therapies are commonly employed for patients with advanced-stage disease [4]. In recent years, immunotherapy has emerged as a promising therapeutic option for select subsets of patients, offering improved clinical outcomes in certain cases [5].

Among immunotherapeutic strategies, immune checkpoint blockade (ICB) has achieved significant success in

treating solid tumors, although the precise mechanisms underlying its efficacy remain a topic of active investigation [6]. A pivotal determinant of ICB effectiveness is the tumor immune microenvironment (TIME), which is traditionally classified into “hot” and “cold” tumors based on the degree of immune cell infiltration [7]. However, whether GC exhibits a distinct binary distribution of hot and cold tumors or represents a continuous spectrum ranging from low to high immune cell infiltration remains contentious [8]. Furthermore, the genetic, epigenetic, and molecular foundations underlying these variations in TIME composition are yet to be fully elucidated.

RNA expression data derived from next-generation sequencing and microarray technologies offer a comprehensive landscape of transcriptomic activity, profiling thousands of transcripts in a single assay. Numerous studies have leveraged this approach to investigate the RNA expression profiles of GC tissues [9]. Among the primary RNA species expressed in cells, mRNA and long non-coding RNAs (lncRNAs) play critical roles in regulating key cellular processes [10, 11]. The wealth of data generated through these analyses provides unprecedented opportunities to unravel the compositional and functional architecture of TIME.

In the present study, we identified two distinct immune phenotypes in GC, categorizing tumors as immune “hot” or “cold” based on their immunological landscapes. A thorough analysis of the genetic, epigenetic, and molecular features associated with these classifications was performed. Furthermore, we defined a gene signature specific to immune hot tumors and developed a robust risk prediction model to stratify the prognosis of GC patients. Notably, we uncovered the pivotal role of the CXCL12–CXCR4 axis in mediating interactions between CD8⁺ T cells and endothelial cells, a hallmark characteristic of immune hot tumors. These findings provided valuable insights into the molecular intricacies of the TIME in GC, offering potential avenues for therapeutic intervention.

Materials and methods

Patients and cohorts

The research workflow is illustrated in Supplementary Fig. 1. RNA expression, DNA methylation data (Illumina Human Methylation 450), DNA copy number variation (CNV), and corresponding clinical information for TCGA–STAD patients were retrieved from the UCSC Xena database, encompassing 373 GC patients [12]. To validate the proposed model, three independent GC datasets, GSE84437, GSE15459, and GSE66229, were obtained from the GEO database.

The GSE84437 dataset is generated using the GPL6947 platform, while the GSE15459 and GSE66229 datasets are based on the GPL570 platform. Expression matrices for these datasets were extracted from the GEO database [13, 14]. Probe mapping was performed using the annotation files corresponding to the GPL6947 and GPL570 platforms, and duplicate probes were averaged to compute the RNA expression levels. The clinical and pathological characteristics of patients in each cohort are summarized in Table S1.

Immune cell infiltration score

The GSVA package was employed to analyze the lncRNA and mRNA expression profiles of TCGA-STAD patients, yielding abundance scores for 28 immune cell types. To ensure the robustness and accuracy of these results, additional analyses were conducted using the XCELL, ESTIMATE, TIMER, quanTIseq, MCP-counter, and EPIC packages [15].

Consensus clustering

Using the immune cell infiltration abundance scores of 28 immune cell types, patient groups were classified through a consensus clustering approach implemented with the ConsensusClusterPlus package. This method employed a consistent clustering-based resampling technique to ensure robust subgrouping. The cumulative distribution function (CDF) curve and proportion of ambiguous clustering (PAC) standard were used as references for determining the optimal number of clusters.

Integrative analysis of mutated genes, DNA methylation, and CNVs

Somatic nonsynonymous point mutations were quantified for each sample using the maftools package, and differences between cold and hot tumor groups were analyzed. Genes with an overall mutation rate > 5%, $P < 0.01$, and an odds ratio (OR) > 1 were identified and screened [16]. To explore differential DNA methylation patterns between cold and hot tumors, the CHAMP package was employed following the imputation of missing values [17]. Type I and Type II probes were analyzed for differences, and differential methylation positions (DMPs) were identified based on adjusted P values < 0.05. Differential methylation regions (DMRs) were determined using the Bumhunter method, with a significance threshold of P values < 0.05. Genes located in CNV regions were annotated using the Genome Research Consortium Human Build 38 (GRCh38) reference genome. Additionally, GenePattern GISTIC_2.0 was utilized for online analysis to identify significant CNV-associated genes.

Weighted gene correlation network analysis (WGCNA)

To construct a scale-free network, the optimal soft-threshold power (β) was calculated. The weighted adjacency matrix was then transformed into a topological overlap matrix (TOM), and the corresponding dissimilarity matrix (1-TOM) was generated. A dynamic tree-cutting algorithm was employed to identify gene modules. To pinpoint RNA modules significantly associated with immune clusters, modules exhibiting the highest correlations were selected for further investigation. Modules were considered top regulators if their module membership (MM) exceeded 0.5 and their gene significance (GS) exceeded 0.45.

Signature construction using machine learning

Genes influencing prognosis were identified through batch univariate Cox regression analysis of the top regulators. Subsequently, 101 machine learning algorithm combinations, integrated with tenfold cross-validation, were employed to develop a robust and accurate predictive model. These algorithms included synthetic approaches such as random survival forests (RSF), elastic net (Enet), Lasso, Ridge regression, stepwise Cox regression, CoxBoost, partial least squares regression for Cox models (plsRcox), supervised principal component analysis (SuperPCA), Gradient Boosting Machine (GBM), and survival support vector machine (Survivor-SVM).

The model's predictive performance was validated using three independent datasets, with the concordance index (C-index) serving as the metric for evaluation. The model demonstrating the highest average C-index across validation datasets was deemed the best-performing predictive model.

Single-cell gene expression quantification and identification of major cell types

A single-cell dataset (GSE183904) was retrieved for analysis [18]. Twelve groups of GC tissue samples (GSM5573467, GSM5573470, GSM5573472, GSM5573475, GSM5573477, GSM5573478, GSM5573480, GSM5573487, GSM5573489, GSM5573491, GSM5573497, and GSM5573501) were imported and processed. Cells were excluded if they contained fewer than 500 molecules, more than 2,500 or fewer than 250 expressed genes, or if over 15% of unique molecular identifiers originated from mitochondrial genes. The FindVariableFeatures function was employed to identify the 2,000 most variable genes for subsequent PCA.

Cells were clustered using the FindNeighbors and FindClusters functions (resolution = 0.2) to delineate cell

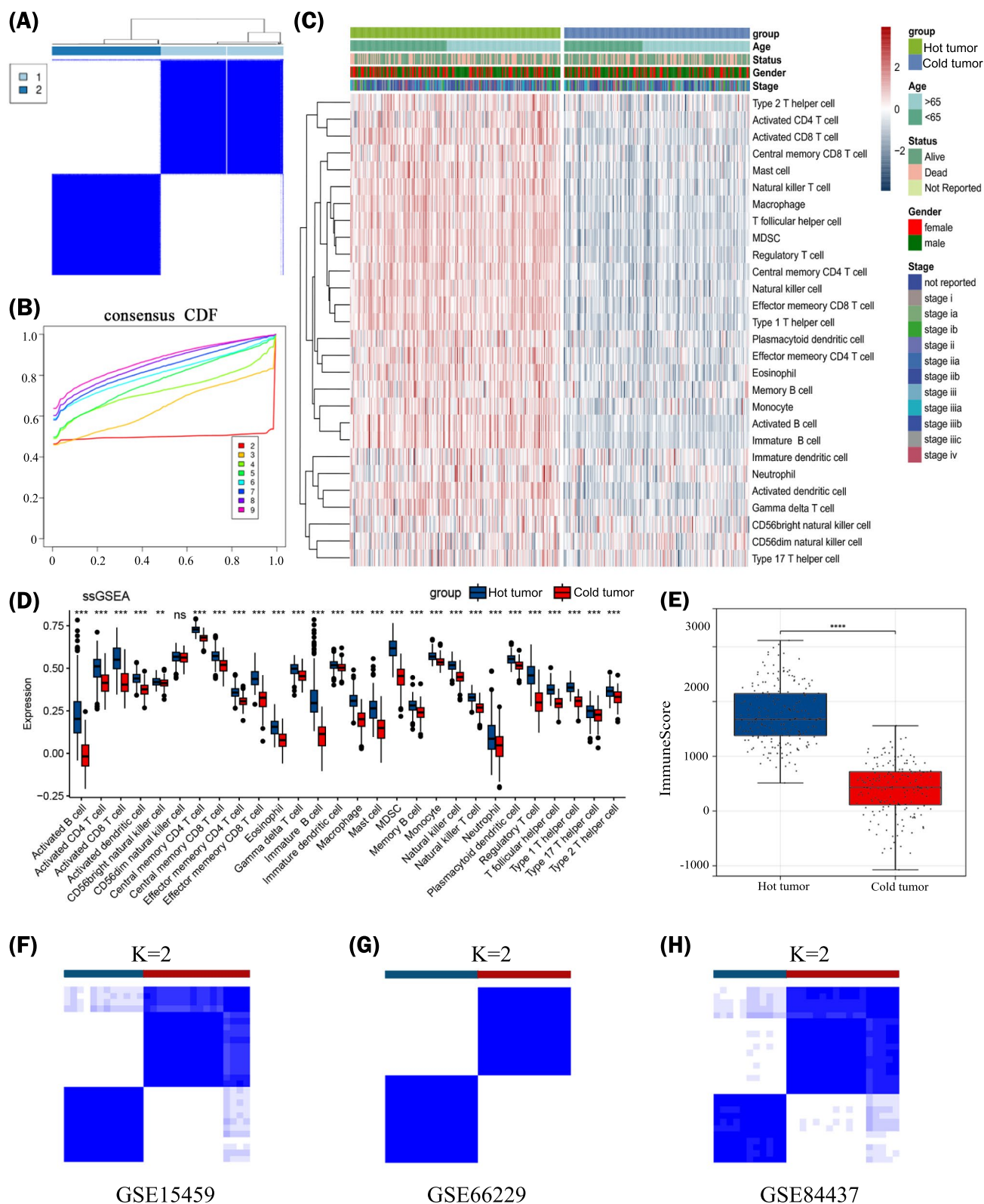


Fig. 1 Identification of cold and hot tumors in GC **A** Consensus score matrix of TCGA-STAD for $k=2$. **B** CDF graph showing consensus distribution for $k=2-9$. **C** Heatmap illustrating the infiltration abundance of 28 immune cell subsets, evaluated using ssGSEA for two clusters. **D** Bar plot comparing the distribution of immune cell infiltration between clusters. Statistical significance: ***, $P<0.001$; **, $P<0.01$; n.s., not significant. **E** Bar plot showing ImmuneScore distribution for TCGA-STAD patients, calculated using ESTIMATE. ****, $P<0.0001$. **F–H** Consensus score matrices for GSE15459, GSE66229, and GSE84437 datasets at $k=2$

subgroups, which were annotated based on established marker genes. The AUCell package was used to compute gene dataset scores with default parameter settings [19].

To investigate the distribution of hot tumor top regulators (HTTR) within specific TIME subpopulations (T cells, NK cells, myeloid cells, and endothelial cells), dimensionality reduction was performed using PCA. Clustering was refined with a graph-based approach (FindClusters function) at a resolution of 0.8 while maintaining default settings. Single-cell trajectories were reconstructed using the Monocle 2 package to capture dynamic changes in TIME cell populations during multi-stage processes [20]. Additionally, the CellChat package was utilized to analyze intercellular communication between immune and stromal cells, with a focus on signaling pathways involving CXCR4 [21].

Immunofluorescence (IF)

Eight GC tissue specimens were obtained via surgical resection. Ethical approval for all human tissue experiments was granted by the Ethics Committee of the Hangzhou First People's Hospital, Westlake University, Zhejiang, China. Written informed consent was obtained from all participants, and the study adhered to the principles outlined in the Declaration of Helsinki.

IF staining was performed on formalin-fixed, paraffin-embedded tissue sections, cut into 5- μ m slides. Slides were deparaffinized and subjected to antigen retrieval by boiling in citrate buffer for 6 min, followed by cooling and washing three times with phosphate-buffered saline for 5 min. Protein blocking was performed with bovine serum albumin for 60 min. Sequentially, BSA blocking, primary and secondary antibody incubations, and antigen retrieval were performed for each antigen, culminating in the addition of three biomarkers: CD45, CD8, and CXCR4.

DAPI was applied as a nuclear stain and incubated for 10 min at room temperature. After washing, an anti-fading mounting medium was used to cover the slides. Fluorescence images were captured using an inverted fluorescence microscope, and IF densities were quantified with ImageJ software.

Statistical analysis

All data processing, statistical analyses, and result visualizations were conducted using R software (version 4.2.2). Categorical variables were compared using either the chi-squared test or Fisher's exact test, depending on the data distribution. For continuous variables, the Wilcoxon rank-sum test or Student's t test was applied as appropriate. Pearson's correlation coefficient was utilized to assess the strength and direction of relationships between two continuous variables. Survival analyses, including Cox proportional hazards regression and Kaplan–Meier survival curves, were performed using the Survival package. Optimal cutoff values for survival analyses were determined with the Survminer package. All statistical tests were two-sided, and a P value of less than 0.05 was considered statistically significant.

Results

A dichotomy of hot and cold tumors in GC

The heterogeneity of GC arises from the differential composition of tumor, stromal, and immune cell types. To dissect the cellular components within GC tissues, we first quantified the abundance of 28 immune cell types across 373 GC samples using single-sample gene set enrichment analysis (ssGSEA). Unsupervised consensus clustering was applied to stratify GC patients into k clusters ($k=2-9$) [15, 22]. Notably, this cohort was optimally separated into two clusters ($k=2$) (Fig. 1A), as determined by the CDF curve of the consensus score matrix, the PAC statistics (Fig. 1B), and so on. The two clusters, designated as C1 and C2, exhibited significant differences in immune cell infiltration. Cluster C1 demonstrated a markedly higher abundance of immune cell infiltration compared to C2 (Fig. 1C and D). Based on these observations, C1 was classified as a “hot tumor” and C2 as a “cold tumor.”

To validate that the two consensus clusters were not biased by the analysis algorithm or the reliance on the TCGA cohort, we employed alternative methods, including xCell, ESTIMATE, TIMER, quanTIseq, MCP-counter, and EPIC, in place of ssGSEA. These analyses consistently corroborated the segregation of GC tissues into hot and cold tumors (Fig. 1E and Supplementary Fig. 2A–E).

To further confirm the robustness of this classification, immune cell infiltration was evaluated in three independent datasets, GSE15459 ($n=192$), GSE66229 ($n=300$), and GSE84437 ($n=433$), followed by unsupervised clustering. Results from these datasets again supported the distinction between hot and cold tumors in GC tissues (Fig. 1F–H). In summary, these findings demonstrated that clinical GC tissues could be reliably stratified into hot and cold tumor

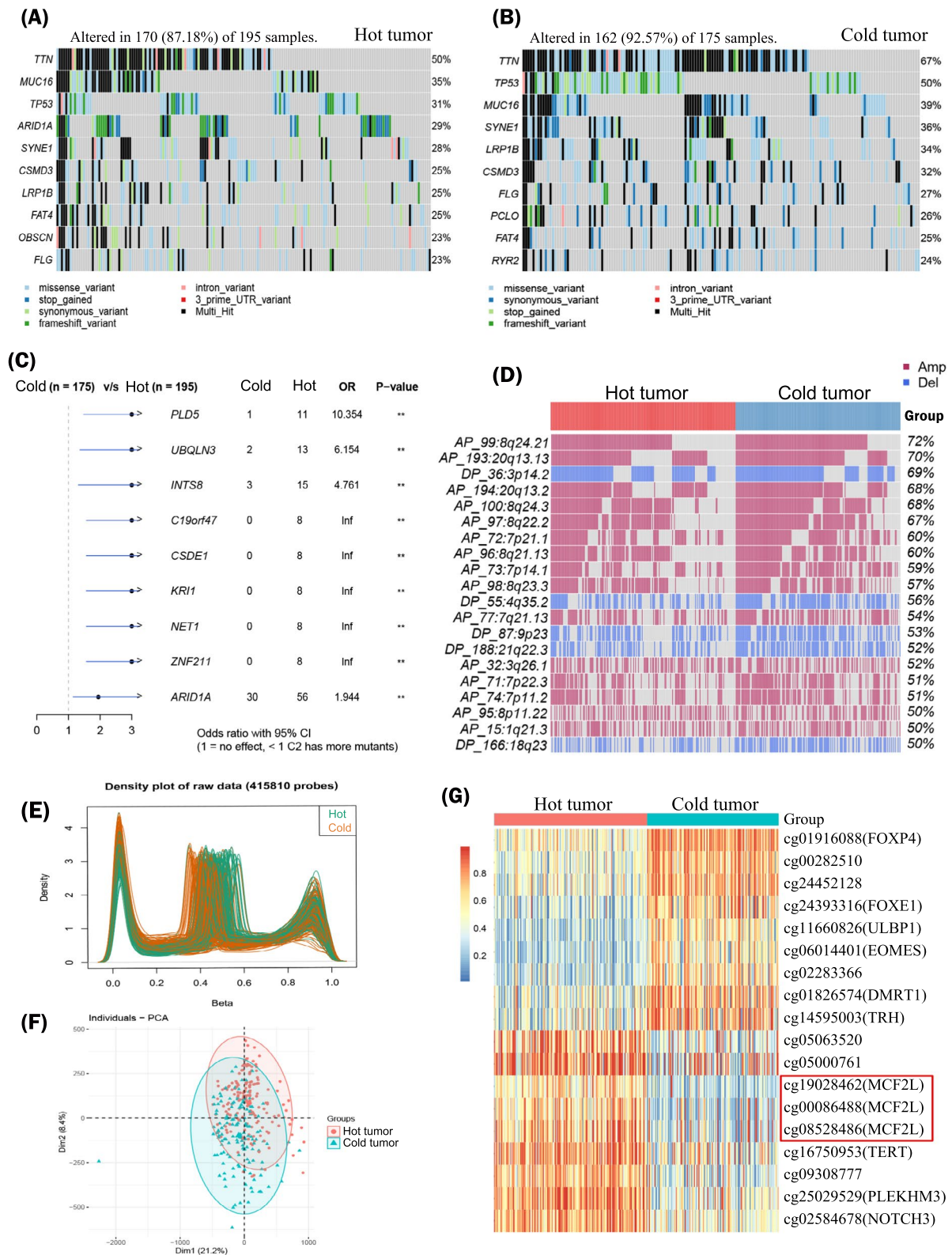


Fig. 2 Comprehensive analysis of immunologically distinct GC tumors **A, B** Oncoplots showing mutation profiles for clusters hot and cold tumor, highlighting the top 10 frequently mutated genes. **C** Forest plot of nine differentially mutated genes between clusters ($P < 0.01$, mutation rate $> 5\%$, $OR > 1$). **D** Comparative oncoplot of significant CNV regions between hot and cold tumor. **E** Density plot of 415,810 probes from TCGA-STAD, stratified by clusters. **F** PCA of normalized methylation position data for hot and cold tumor. **G** Heatmap showing DMPs and associated genes between clusters ($|\log_{10}fc| > 0.25$)

categories, providing a foundation for further investigation into the biological and clinical significance of this dichotomy.

Comprehensive analysis of immunologically distinct GC tumors

Analysis of clinicopathological parameters revealed that hot tumors were significantly associated with a higher proportion of poorly differentiated cancer cells (G3) and more advanced tumor stages (T3) (Supplementary Fig. 3A, B), suggesting a potential link between cancer cell stemness and the TIME. Conversely, no significant differences were observed between hot and cold tumors in terms of Barrett's esophagus prevalence, *H. pylori* infection, metastatic status, or overall survival (OS) time (Supplementary Fig. 3C–I, Table S2).

The distinctive TIME may result from underlying genetic and epigenetic alterations intrinsic to cancer cells. To investigate this, we compared single nucleotide polymorphisms (SNPs), DNA methylation profiles, and CNVs between hot and cold tumors. A distinct mutation landscape was evident between the two groups. While both groups shared common mutations such as *MUC16*, *SYNE1*, and *LRP1B* (Fig. 2A, B), certain mutations were preferentially enriched. Hot tumors exhibited a higher prevalence of *ARID1A* mutations ($OR = 1.94$; $P < 0.01$) and fewer *TTN* ($OR = 0.50$; $P < 0.01$) and *TP53* mutations ($OR = 0.46$; $P < 0.001$) (Fig. 2C, Table S3).

To identify genes specifically enriched in hot tumors, we screened for mutations with $P < 0.05$ and $OR > 1$, yielding 69 genes significantly enriched in pathways including herpes simplex virus 1 (HSV-1) infection, liver cancer, colorectal cancer, and EGFR tyrosine kinase inhibitor resistance (Supplementary Fig. 4A). In contrast, cold tumors showed enrichment for 219 genes ($P < 0.05$, $OR < 1$), with prominent involvement in HSV-1 infection, glioma, and amino sugar/nucleotide sugar metabolism pathways (Supplementary Fig. 4B).

Analysis of CNVs revealed high mutation rates in regions such as 8q24.21 and 20q13.13, exceeding 70% (Fig. 2D). Both tumor types shared common amplification sites, including 5q11.2 and 1q21.3. However, cold tumors exhibited unique deletions at 3p14.2 and 4q13.2, as well as amplifications at 17q21.2. In hot tumors, significant

amplifications were identified at 3q29, 6p21.32, and 20p13 (Supplementary Fig. 4C). Notably, certain chromosomes in hot tumors displayed consistent variation across the entire chromosome (e.g., amplification of chromosomes 7, 13, and 20, and loss of chromosome 4). Other chromosomes, such as 1, 3, and 5, exhibited contrasting CNV patterns at the arm level (Supplementary Fig. 4D, Table S4). For instance, chromosome 5q showed substantial copy number amplification, while chromosome 5p did not.

In terms of DNA methylation, hot tumors demonstrated higher overall methylation levels, as visualized by density maps (Fig. 2E). PCA revealed that TCGA-STAD patients could be distinctly separated into two groups based on methylation profiles (Fig. 2F). Differential analysis identified nine upregulated and nine downregulated methylation sites in hot tumors, with three highly significant sites (cg19028462, cg00086488, and cg08528486) located within the *MCF2L* gene body (Fig. 2G, Supplementary Fig. 4E, Table S5). Interestingly, *MCF2L* expression was significantly downregulated in hot tumors (Supplementary Fig. 4F).

DMRs also exhibited variation in GC. Functionally, the methylation degree was higher in gene bodies, TSS1500, and TSS200 regions compared to the 1st exon region. Based on CpG content and neighborhood context, methylation levels in CpG shores were higher than in CpG islands (Supplementary Fig. 4G) [23].

Identification of a hot tumor-related gene module

We hypothesized that specific gene modules played pivotal roles in the formation of hot tumors. To investigate this, we employed WGCNA to identify gene modules associated with clinical GC tissues. A soft-threshold power of 4 was selected based on the scale-free topology fit index (Supplementary Fig. 5A). This analysis identified 10 distinct gene modules within the 373 GC tissues (Fig. 3A), with eigengene adjacency relationships visualized in Supplementary Fig. 5B.

Correlation analysis revealed a strong association between the red gene module and hot tumors ($cor = 0.76$) (Fig. 3B). This module comprised 1,769 genes, including cytokines such as *IFNG*, *CCL3*, and *CCL7*; cell type markers like *CD3D*, *CD4*, *CD8B*, *NKG7*, *CD14*, and *LYZ*; and immune regulatory molecules such as *LAG3*, *ICOS*, and *CTLA4*. These components underscore the red module as a hot immune-related gene cluster (Table S6).

To identify key regulators within the red module, we analyzed the MM, reflecting the importance of a gene within the module, and GS, indicating its relationship with phenotypic traits. Filtering genes with $MM > 0.5$ and $GS > 0.45$ yielded 446 genes as top regulators within the red module (Fig. 3C).

Pathway enrichment analysis using KEGG revealed that these top regulators were significantly involved in cytokine-cytokine receptor interactions and chemokine

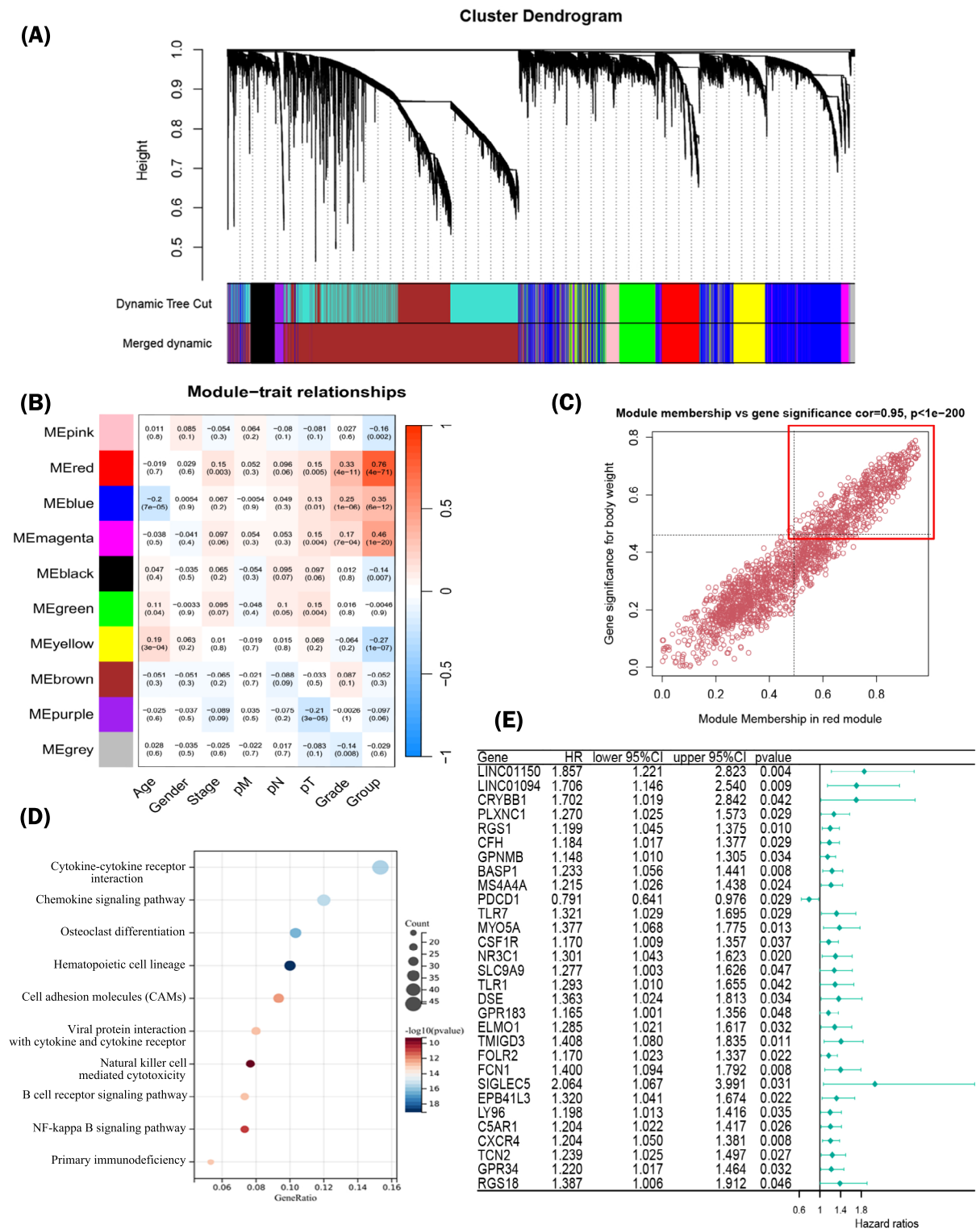


Fig. 3 Identification of a hot tumor-related gene module **A** Cluster dendrogram of differentially expressed genes based on TOM dissimilarity. **B** Heatmap of correlations between modular eigengenes and clinical traits in TCGA-STAD. **C** Scatter plot showing a high correlation

between GS and MM in the red module ($cor=0.95$). Top regulators (GS > 0.45, MM > 0.5) are highlighted. **D** Bubble plot of KEGG pathway enrichment analysis for top regulators. **E** Forest plot of univariate Cox analysis for 30 prognostic RNAs in top regulators

signaling pathways, highlighting active immune crosstalk within the TIME (Fig. 3D, Table S7). Furthermore, Cox proportional hazards regression analysis identified 30 genes significantly associated patient survival in GC (Fig. 3E). In conclusion, this analysis identified a robust gene set underpinning the formation and immune characteristics of hot tumors in GC, providing valuable insights into the molecular basis of tumor immunity.

Machine learning-based construction of a prognostic prediction model

To establish a risk model for predicting patient survival, we focused on the 30 genes identified in the previous analysis. After excluding *LNC01150* and *TMIGD3*, genes that were not detectable in GC tissues across other datasets, we finalized a set of 28 genes, which we termed HTTR. Given the robust capabilities of machine learning algorithms in classifying patients based on survival time, we fitted 101 prediction models to the TCGA-STAD dataset. Among these, the RSF model outperformed others, achieving a C-index of 0.87.

We further validated the RSF model by testing its performance on three independent GC datasets, GSE15459, GSE84437, and GSE66229. The C-index values for these datasets were 0.57, 0.53, and 0.59, respectively, with RSF consistently producing the highest average C-index across all datasets (Fig. 4A, Table S8). The final prediction model comprised 236 decision trees. Variable importance analysis revealed that the top three genes contributing to the model were *BASPI1*, *PDCD1*, and *TCN2* (Fig. 4B).

To assess the predictive performance, we applied the model to four datasets, and the results demonstrated a clear stratification of patients based on survival time across all datasets (Fig. 4C–F). In the TCGA cohort, the optimal cut-off for HTTR was 49.70099, with a median of 44.12556 and a range of 16.91484 to 120.77760. To evaluate its specificity and sensitivity in predicting survival at 1, 3, and 5 years, we performed a receiver operating characteristic (ROC) analysis. In the TCGA dataset, the model achieved an area under the curve (AUC) > 0.9, while the AUC values for the other three datasets were around 0.6 (Supplementary Fig. 5C–F).

Next, we compared the performance of our prediction model with other key clinical parameters through univariate and multivariate analyses (Supplementary Fig. 6A–H). Our model was found to be an independent risk factor for patient survival, even though its prediction efficiency was relatively lower in the three GEO cohorts, consistent with the results from the ROC analysis.

Comprehensive evaluation of the relationship between model score and immunological factors

To what extent does our model score reflect the immune landscape of GC? We investigated this question by analyzing the correlation between model scores and three key aspects of immunological characteristics (Table S9).

First, we observed significant differences in immune cell types between the high- and low-score groups (Supplementary Fig. 7A, B). Notably, 24 of the 28 deconvoluted immune cell types exhibited a positive correlation with the model score (Supplementary Fig. 7C). Applying a more stringent criterion ($R > 0.4$, $P < 0.001$), we identified 21 immune cell types with strong correlations, with effector memory CD4 T cells and mast cells emerging as the top two most correlated immune cells.

Second, we explored the role of chemokines, essential mediators of immune processes, and their receptors. Our analysis revealed significant correlations between the model score and nine chemokines as well as eight chemokine receptors. Among these, *CCL14*, *CXCL12*, *CCR1*, and *CXCR4* were the most strongly correlated (Supplementary Fig. 7D).

Finally, we examined the relationship between the model score and 38 membrane-localized immunoregulatory molecules, which encompass functional categories such as immune checkpoints (e.g., *CD27* and *CD48*), antigen presentation molecules (e.g., *HLA* family), and immunoinhibitors (e.g., *KDR* and *TGFBRI*). Of these, 23 molecules showed significant correlations with the model score, with *CXCR4* and *CXCL12* ranking as the top two (Supplementary Fig. 7E). This finding was particularly intriguing, as *CXCL12* serves as the cognate ligand for *CXCR4*, indicating a robust activation of the CXCL12-CXCR4 signaling axis underlying the model score. The CXCL12-CXCR4 axis is well established in the literature for its role in recruiting *CXCR4*-expressing progenitor hematopoietic and immune cells to sites of angiogenesis. In summary, our findings demonstrated a strong association between the model score and the TIME of GC, further highlighting the biological relevance and utility of the predictive model in characterizing tumor-immune interactions.

Cellular network underlying HTTR

HTTR represents a cluster of master regulators governing diverse cellular programs across various cell types in hot tumors. Quality-controlled single-cell data comprising 30,000 cells were unsupervisedly clustered into 10 compartments. Eight major cell types were identified: T cells, NK cells, B cells, epithelial cells, myeloid cells, fibroblasts, mast cells, and endothelial cells (Table S10). The overall activity of HTTR was predominantly observed in T cells, NK cells, and myeloid cells (Fig. 5A, B). Among the

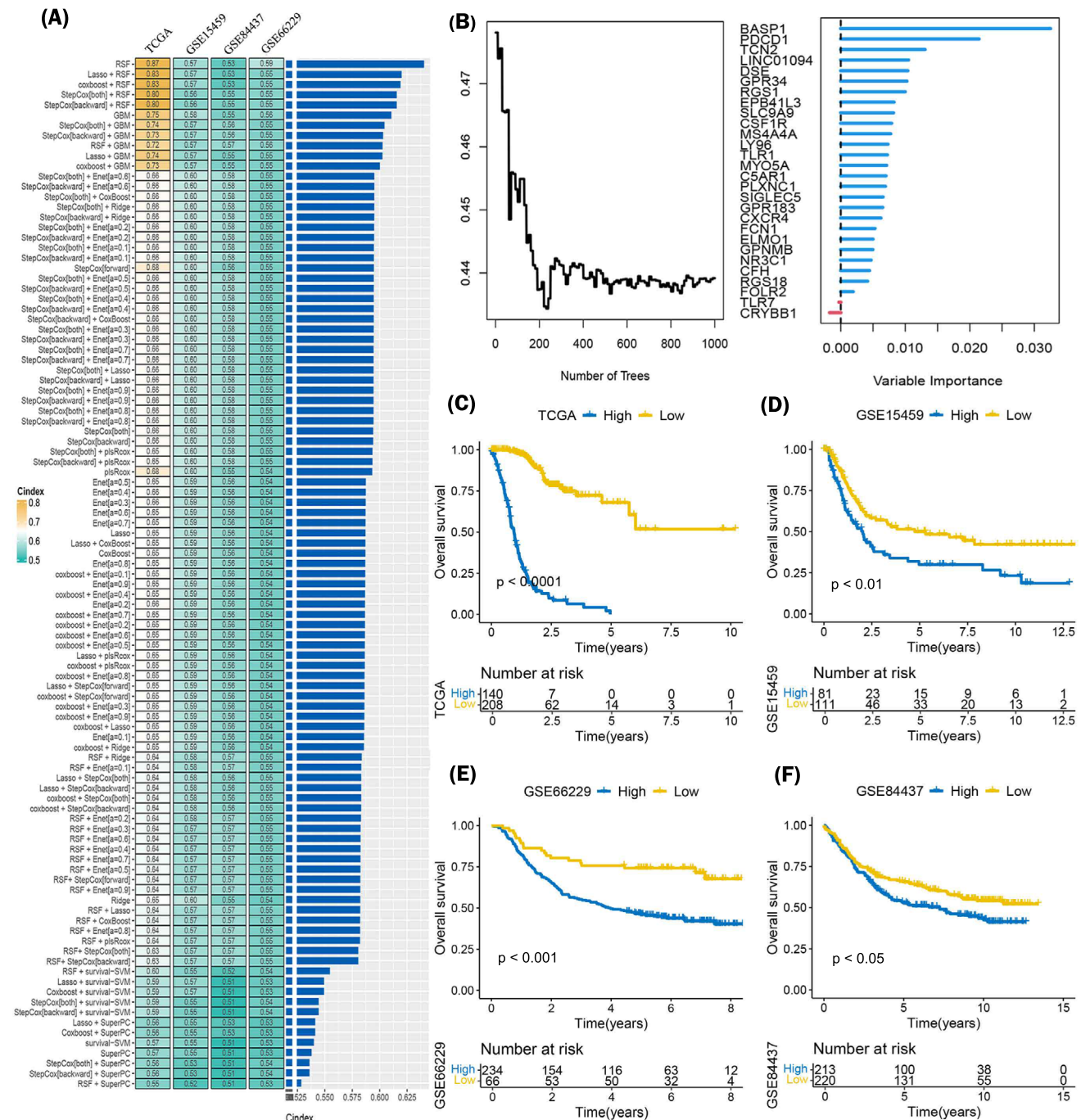


Fig. 4 Machine learning-based construction of a prognostic prediction model **A** Performance of 101 prediction models using tenfold cross-validation and C-index evaluation. **B** Bar plot showing the

importance of the 28 HTTR genes in the RSF model. **C–F** Kaplan-Meier curves of OS for high- and low-risk groups in TCGA (**C**), GSE15459 (**D**), GSE66229 (**E**), and GSE84437 (**F**)

genes, *CXCR4* was primarily expressed in T and NK cells, while *C5AR1* and *BASP1* were predominantly expressed in myeloid cells (Fig. 5C). Notably, *CXCR4* emerged as the most highly expressed gene within the HTTR cluster (Supplementary Fig. 8).

Intercellular communication analysis revealed that the increased activity of the CXCL signaling pathway was largely attributed to the heightened expression of the CXCL12-CXCR4 axis (Fig. 5D). Moreover, endothelial cells exhibited the strongest interaction with T cells within

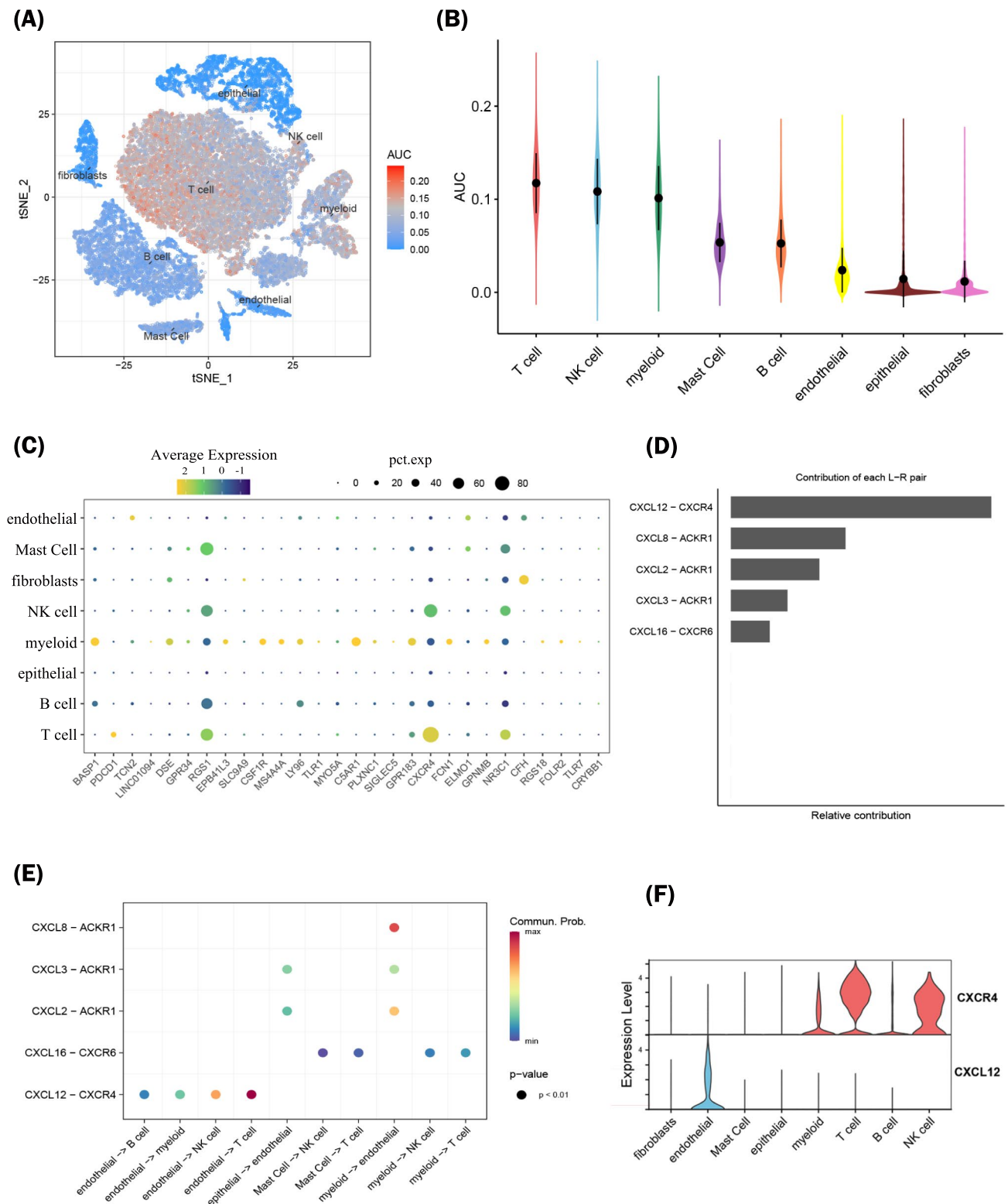


Fig. 5 Cellular network of HTTR in preliminary clustering **A** T-SNE plot showing AUCell score distribution across cells. **B** Violin plot of AUCell score concentration in each cell type. **C** Bubble plot of HTTR gene expression in individual cells. **D** Bar plot showing the contribu-

tion of ligand-receptor pairs in CXCL signaling pathways. **E** Bubble plot of significant interactions (ligand-receptor pairs) in CXCL signaling pathways. **F** Violin plot of *CXCR4* and *CXCL12* expression in all cells post-clustering

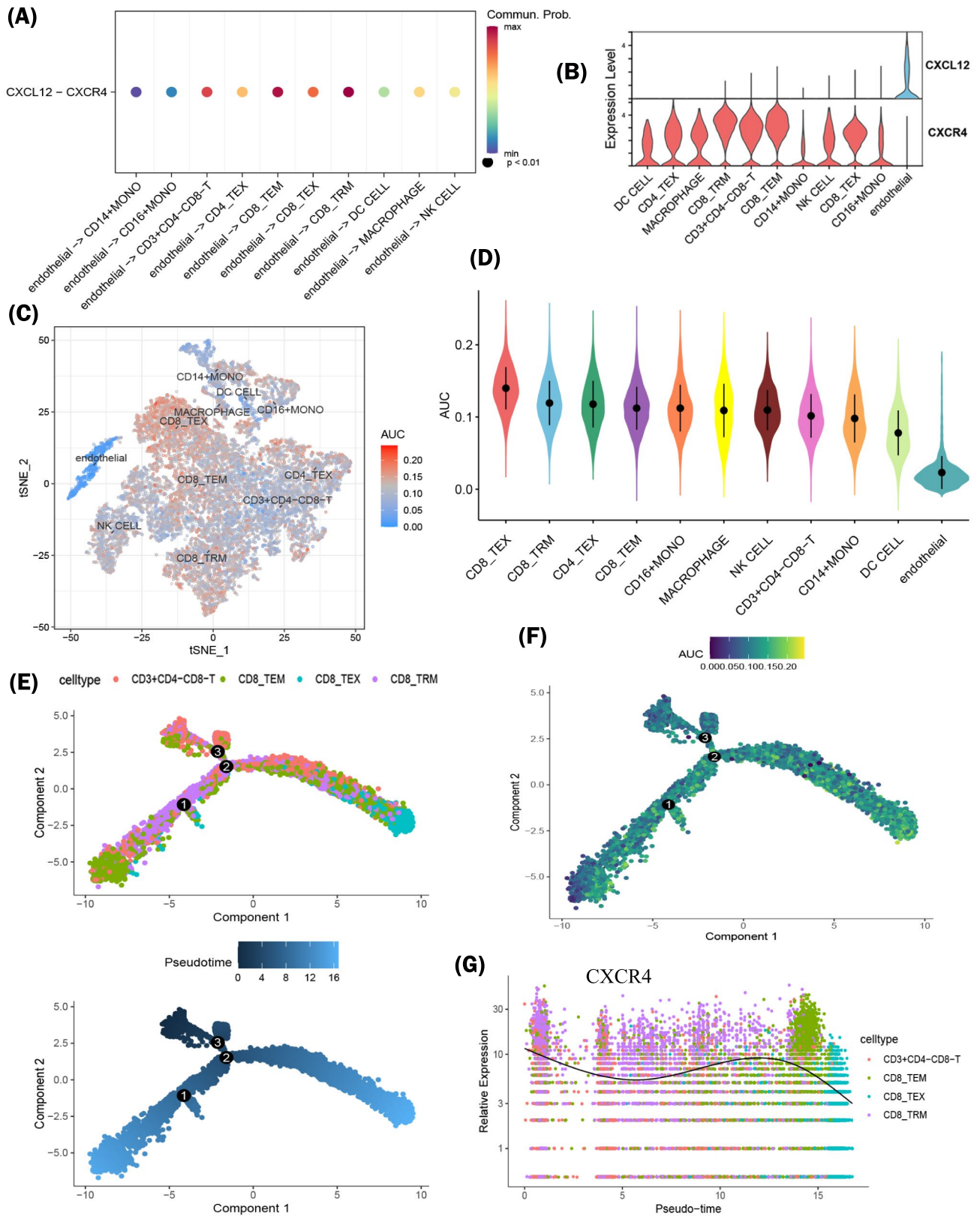


Fig. 6 Cellular network of HTTR in secondary clustering **A** Bubble plot showing CXCL12-CXCR4 axis interaction strength in the second cluster. **B** Violin plot of *CXCR4* and *CXCL12* expression in the second cluster. **C** T-SNE plot showing AUCell score distribution post-secondary clustering. **D** Violin plot of AUCell score concentration post-secondary clustering. **E–G** Pseudotemporal trajectory analysis of CD8+T cells, showing gradient pseudotime ordering, AUCell score dynamics, and relative *CXCR4* expression

the CXCL12-CXCR4 axis, with *CXCL12* being highly expressed in endothelial cells (Fig. 5E, F). These findings underscored the pivotal role of the CXCL12-CXCR4 axis, primarily mediating interactions between endothelial cells and immune cells.

To investigate the CXCL12-CXCR4 axis in greater detail, we further grouped T cells, NK cells, myeloid cells, and endothelial cells into clusters. Eleven major cell types were identified, including CD8+TEX, CD8+TEM, CD8+TRM, CD4+TEX, CD16+MONO, CD14+MONO, NK cells, CD3+CD8-CD4-T cells, macrophages, dendritic cells, and endothelial cells. *CXCR4* exhibited the highest expression in the second cluster (Supplementary Fig. 9). Subsequent analysis of interactions within the CXCL12-CXCR4 axis revealed that CD8+T cells (including CD8+TEX, CD8+TEM, CD8+TRM, and CD3+CD8-CD4-T) demonstrated strong interactions with endothelial cells, with CD8+TRM showing the strongest interaction (Fig. 6A). *CXCR4* was most strongly expressed in CD8+T cells (Fig. 6B).

AUCell analysis demonstrated that HTTR activity was broadly distributed across immune cells, with the highest activity concentrated in CD8+TEX cells (Fig. 6C, D). Pseudotime analysis further revealed that as CD8+T cells progressed through differentiation, HTTR activity increased overall, while *CXCR4* expression exhibited a declining trend, reaching relatively low levels during the transition into CD8+TRM and CD8+TEX states (Fig. 6E–G).

Finally, we assessed *CD8* expression in GC tumor tissues using IF. The results clearly distinguished GC patients into cold and hot tumor groups based on *CD8* IF density (Fig. 7A). Interestingly, the IF density of *CXCR4* was significantly higher in cold tumors than in hot tumors (Fig. 7B, C).

Discussion

The TIME plays a pivotal role in the onset, progression, and prognosis of GC. In recent years, the rapid advancements in immunotherapy have underscored the importance of comprehensively investigating GC immune subtypes. Such efforts not only elucidate the dynamic changes within the TIME but also identify novel therapeutic targets, facilitate the evaluation of patient prognosis, and enable the development of personalized treatment strategies [24]. Despite

these advancements, the clinicopathological and molecular characteristics of immune hot tumors remain inadequately characterized. To address this knowledge gap, we investigated the relationship between hot tumor-associated genes and GC immunity, prognosis, and molecular interactions. This study provided a deeper understanding of the immune mechanisms underpinning hot tumors, offering valuable insights for immunotherapeutic innovation and precision medicine.

Our results preliminarily conclude that GC tissues can be divided into cold and hot tumors. Although $k=5$ is probably the optimal solution for classification in consensus clustering analysis, however, our preliminary general classification provides a better understanding of the overall differences between hot and cold gastric cancer. In one way, this provides a solid basis for a more detailed analysis to follow. Notably, *H. pylori* infection did not influence whether a tumor was classified as hot or cold. This finding aligned with observations that *H. pylori* infection is not a biomarker for immunotherapy efficacy in GC but instead diminishes its effectiveness through immunosuppressive mechanisms [25].

Gene enrichment analysis revealed that hot tumors harbored more mutations in genes associated with signal transduction pathways, whereas cold tumors exhibited mutations predominantly linked to metabolic pathways and malignant progression. Intriguingly, both tumor types were enriched in the HSV-1 infection pathway after initial classification. While the direct causal relationship between HSV-1 infection and GC remains unclear, prior studies have suggested a positive association between active gastric HSV-1 infection and functional gastrointestinal disorders [26]. This raises the possibility that HSV-1 infection may contribute to the development of gastrointestinal symptoms.

At the genetic level, our findings highlighted *ARID1A* as a key mutation significantly associated with the hot tumor subtype of GC. Conversely, mutations in *TP53* and *TTN* were more prominent in cold tumors. Previous studies have shown that *ARID1A* mutations predict sensitivity to immune checkpoint inhibitors across various cancers [27, 28], while *TP53* mutations are known to suppress immune responses in GC [29]. These findings suggest that high *ARID1A* mutation rates and low *TP53* mutation rates may be defining features of the hot immune response in GC. Additionally, *TTN* mutations may hold potential as predictive markers for immune responses; however, further research is needed to validate this hypothesis.

CNVs also influence gene expression and were evident in both tumor subtypes. Notably, significant amplifications were observed at the 5q11.2 and 1q21.3 loci in both hot and cold tumors. Previous research has established a strong association between the 5q11.2 locus and the prognosis of GC patients, further underscoring its clinical significance [30]. In cold tumors, our research identified significant

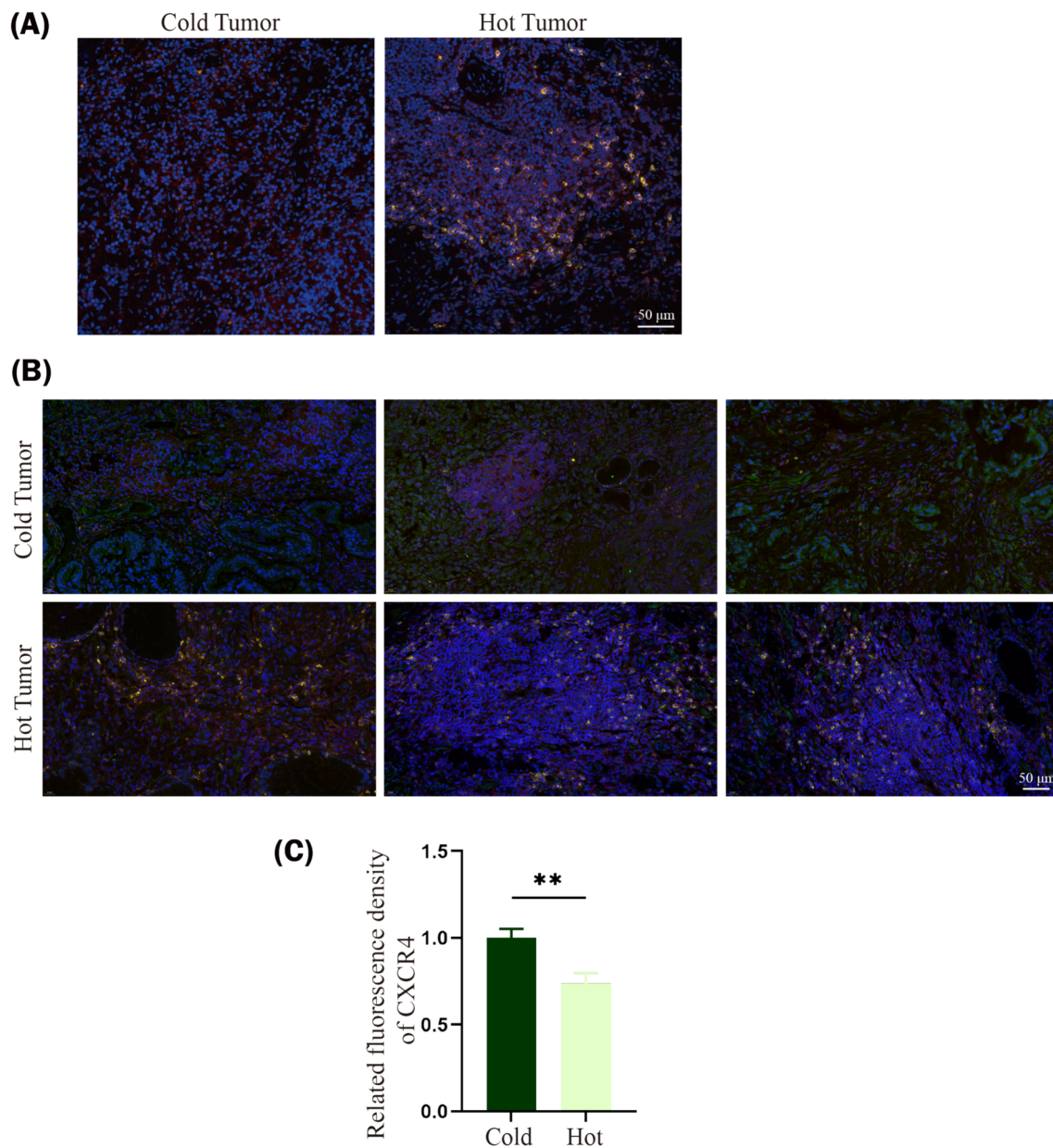


Fig. 7 IF analysis of CXCR4 in GC tumors **A** IF images of *CD8* (yellow) and *CD45* (red) in hot and cold tumors. **B** IF images of *CD8*, *CD45*, and *CXCR4* (green) in hot and cold tumors. **C** Bar plot showing significantly higher *CXCR4* IF density in cold tumors compared to hot tumors

deletions at the 3p14.2 and 4q13.2 loci and amplifications at the 17q21.2 locus. Notably, loss of heterozygosity (LOH) at the 3p14.2 locus has been frequently observed in early-stage GC, suggesting its potential role in early tumorigenesis [31]. In contrast, hot tumors exhibited significant amplifications at the 3q29, 6p21.32, and 20p13 loci. These distinct CNV patterns highlighted critical differences between hot and cold tumors, offering valuable insights into their respective roles in tumor development and progression.

DNA methylation, a key regulator of gene expression, significantly influences interactions between tumor and immune

cells [32]. In hot tumors, *MCF2L* methylation remained stable, potentially contributing to the formation of the hot tumor subtype. However, the precise role of *MCF2L* methylation in shaping the hot tumor subtype remains unclear and warrants further investigation.

Using WGCNA, we identified gene modules most strongly correlated with hot tumors, screened for genes impacting prognosis, and developed a consensus HTTR signature using machine learning. By fitting 101 models with tenfold cross-validation across four independent datasets, we determined that the RSF model was optimal.

Genes contributing significantly to the RSF model showed strong associations with immune cell infiltration. For example, *BASPI*, a transcriptional co-repressor interacting with oncogenic transcription factors, has been closely linked to immune cell infiltration in GC [33]. *PDCD1*, an inhibitory receptor expressed by activated T cells, plays a central role in ICB therapy and has been one of the most successful therapeutic targets [34]. Conversely, the role of *TCN2* in GC remains underexplored. Although *TCN2* is known to mediate cellular vitamin B12 import, its potential involvement in the GC TIME requires further investigation.

ROC analysis demonstrated that the HTTR signature reliably predicted 1-year, 3-year, and 5-year OS in GC patients, with high accuracy. Furthermore, the HTTR signature emerged as an independent prognostic factor, outperforming traditional demographic predictors such as age and sex. Importantly, patients with higher HTTR scores were associated with poorer survival outcomes.

Although traditional pathological staging retains robust prognostic value, integrating the HTTR signature with staging has the potential to enhance patient stratification. Future efforts will focus on combining the HTTR signature with pathological staging to develop a dual-index stratification system, thereby optimizing its clinical applicability and improving personalized treatment strategies.

Single-cell-level analysis provides a more precise characterization of the TIME [35]. Our study revealed that HTTR was predominantly enriched in T cells, NK cells, and myeloid cells, aligning with an “immune-hot” phenotype [36]. Further investigation of individual genes within the HTTR cluster demonstrated strong activity of *CXCR4*, *C5AR1*, and *BASPI* in these immune cell populations [33, 37]. Among these, *CXCR4* emerged as the most widely expressed gene within HTTR and showed a high correlation with the risk score in our model’s analysis of immune factors.

CXCR4, a rhodopsin-like G protein-coupled chemokine receptor (GPCR), is well documented for its role in cancer biology [38]. Previous studies have shown that *CXCL12*, secreted by multiple myeloma cells, attracts *CXCR4*-expressing monocytes into the tumor niche, inducing their differentiation into M2 macrophages, which in turn support tumor growth [39]. Similarly, in osteosarcoma, the *CXCL12/CXCR4* axis facilitates the migration and survival of myeloid-derived suppressor cells while suppressing cytotoxic T cell (CTL) expansion, thereby promoting tumor progression [40]. Furthermore, in prostate cancer, *CXCR4* inhibitors effectively suppress endothelial cell proliferation and migration [41]. These findings suggest that the *CXCR4*-mediated signaling pathways may play a significant role in the TIME of GC.

Our cell communication analysis highlighted that the *CXCL12-CXCR4* axis within the *CXCL* signaling pathway generated the strongest intercellular signals. Specifically,

CXCL12, predominantly expressed in endothelial cells, exhibited its strongest interactions with *CXCR4*, which is primarily expressed in T cells. These findings underscored the pivotal role of the *CXCL12-CXCR4* axis in mediating interactions between endothelial cells and immune cells within the GC TIME.

When we examined *CXCR4* expression in immune cell subpopulations, both the overall and individual gene activities of HTTR were strongly associated with CD8+ T cells, with the most intense interactions occurring between endothelial cells and CD8+ TRM cells. Pseudotime analysis revealed that while *CXCR4* expression decreased during T cell differentiation, the overall activity of HTTR increased. Notably, *CXCR4* expression is relatively low in CD8+ TRM and CD8+ TEX cells [42, 43]. This suggested that tumor endothelial cells interacted with CD8+ TRM and CD8+ TEX cells through the *CXCL12-CXCR4* axis, inducing a robust immune response despite low *CXCR4* expression. This conclusion was consistent with previous findings showing that *CXCR4* blockade reduces breast cancer metastasis, increases T lymphocyte infiltration, and enhances the efficacy of ICB therapy [44]. Additionally, emerging evidence indicates that inhibiting *CXCR4* reduces metastasis in GC [45]. Our IF results suggest that low *CXCR4* fluorescence density seems to be more dominant in hot tumors, but more samples need to be included for validation. These findings provided novel insights into the TIME of GC and the mechanisms by which immune cells exerted their influence on tumor progression.

However, this study has several limitations. First, all data analyzed in this study were derived from single-center retrospective studies. To confirm the clinical utility and generalizability of the HTTR, future research should incorporate multi-center datasets. Second, the imbalance in clinical information and gene expression data from public repositories might have introduced bias into the model. Addressing this issue with more uniform and comprehensive datasets will enhance model reliability. Third, further investigations are required to elucidate the dynamic interactions between immune cells and endothelial cells, particularly their functional states in both spatial and temporal contexts. Finally, while this study broadly categorized GC into hot and cold tumors, the TIME is influenced by numerous factors. Future research should aim to refine these classifications by incorporating additional covariates and employing more detailed subgrouping strategies. Such efforts will provide a more nuanced and comprehensive understanding of the tumor immune landscape.

Conclusions

GC was preliminarily categorized into hot and cold tumors, which exhibited distinct genetic, epigenetic, and molecular characteristics. Building on these findings, we developed

the HTTR signature, comprising genes strongly associated with hot tumors. As an independent influence, HTTR was of some value in predicting the survival prognosis of gastric cancer patients. In the future, HTTR in combination with staging diagnosis may be a more individualized option for gastric cancer patients. Additionally, our data highlighted the critical role of the CXCL12-CXCR4 axis in mediating interactions between endothelial cells and CD8⁺ T cells. Notably, CD8⁺ T cells with relatively low CXCR4 expression appeared more prone to engaging with endothelial cells, further emphasizing the significance of this axis in the TIME.

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Data availability Raw RNA-seq, microarray, and scRNA-seq data are available in the public open access repository (NCBI database). All data analyzed in this study are included in this manuscript.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this study.

Ethical approval and consent to participate Not applicable.

Consent for publication Not applicable.

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